Mathematics - Graduate Level Assessment 24 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. How many integers between 1 (included) and 100 (included) are divisible by either 2, 3, or 5?

   (a) 82
   (b) 60
   (c) 80
   (d) 68
   (e) 74

2. For how many positive integers k does the ordinary decimal representation of the integer k! end in exactly 99 zeros?

   (a) Twenty
   (b) Three
   (c) None
   (d) Four
   (e) Five

3. For a$\geq$ 0, we define $S_a = x|dist(x, S) \leq a$, where $dist(x, S) = inf_{y \in S}||x - y||$. Suppose S is convex. Is S_a convex? Return 1 for yes and 0 for no.

   (a) Undefined
   (b) 0.0
   (c) None of the above
   (d) 1.0

4. At West Elementary School, there are 20 more girls than boys. If there are 180 girls, how can you find the number of boys?

   (a) divide 20 by 180
   (b) add 180 to 20
   (c) multiply 180 by 20
   (d) multiply 180 by 2
   (e) divide 180 by 20

5. We know that $y' = (x+y)/2$, we also know that $y(x = 0) = 2, y(x = 0.5) = 2.636, y(x = 1) = 3.595, y(x = 1.5) = 4.9868$, what is the value of y(2) using Adams bashforth predictor method.

(a) 5.4321
(b) 7.1234
(c) 6.2456
(d) 7.7654
(e) 5.6789

True / False (10 marks)

Answer True or False

6. True or False: The value of the expression -16 + 13 + (-33) is equal to -36.

7. True or False: The correct answer to 'Given that $n > 1$, what is the smallest positive integer n whose positive divisors have a product of n^6?' is '120'.

8. True or False: There is only one group of order n (up to isomorphism) for the integers n = 4, 6, 8, and 10.

9. True or False: The probability that the sample mean lies between 74.2 and 78.4 is approximately 0.6398 for the given normal distribution.

10. True or False: The composition f(g(h(x))) yields the expression -225 x^2 + 90 x - 27, which is a common miscalculation in function composition.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the relationship between reading rate and time. If Susan reads a book at a rate of 1 page every 3 minutes, explain how to calculate the total time required for her to read 18 pages.

12. Analyze the impact of a 15% increase in waist circumference on Owen's current jean size of 32 inches. Discuss the mathematical calculations involved and the implications for clothing sizing.

End of Paper